Supplementary Online Content

Daoust R, Paquet J, Marquis M, et al. Evaluation of interventions to reduce opioid prescribing for patients discharged from the emergency department: a systematic review and meta-analysis. *JAMA Netw Open*. 2022;5(1):e2143425. doi:10.1001/jamanetworkopen.2021.43425

eMethods 1. Detailed Search Strategy and Search Term Alternatives for Each Database

eMethods 2. Description of the Interventions Design to Reduce Opioid Prescription and How They Were Categorized

eFigure 1. Example of the 6-mo Step Changes Representation From Interrupted Time Series Data

eFigure 2. Risk of Bias Assessed With the ROBINS-I Tool for ITS and Preintervention-Postintervention Study Designs

eFigure 3. Risk of Bias Assessed With the EPOC Risk of Bias Tool for the RCT and Cohort Study Designs

eFigure 4. Forest Plots of Opioid Prescription Rate in RCT, Preintervention-Postintervention, and Cohort Studies

eFigure 5. Forest Plots of Opioid Prescription Quantity in RCT, Preintervention-Postintervention, and Cohort Studies

eFigure 6. Forest Plot of the 1-y Step Change in Opioid Prescription Rate for ITS Studies

eFigure 7. Forest Plot of the 1-y Step Change in Prescribed Opioid Quantity for ITS Studies

eFigure 8. Forest Plot of the 6-mo Step Change in Opioid Prescription Rate for ITS Studies Excluding Studies at High Risk of Bias

eFigure 9. Forest Plot of the 6-mo Step Change in Prescribed Opioid Quantity for ITS Studies Excluding Studies at High Risk of Bias

eFigure 10. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Opioid Prescription Rate for ITS Studies

eFigure 11. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Opioid Prescription Rate for RCT, Preintervention-Postintervention, and Cohort Studies

eFigure 12. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Prescribed Opioid Quantity for ITS Studies

eFigure 13. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Prescribed Opioid Quantity for RCT and Preintervention-Postintervention Studies

eReferences

This supplementary material has been provided by the authors to give readers additional information about their work.

© 2022 Daoust R et al. *JAMA Network Open*.

eMethods 1. Detailed Search Strategy and Search Term Alternatives for Each Database

Search Strategy

Ovid MEDLINE

1. exp Narcotics/	
2. exp Opiate Alkaloids/	
3. opioid*.mp.	
4. opiate*.mp.	
5. opium*.mp.	
6. narcotic*.mp.	
7. alfentanil.mp.	
8. buprenorphine.mp.	
9. butorphanol.mp.	
10. codeine.mp.	
11. dextropropoxyphene.mp.	
12. fentan?l.mp.	
13. hydrocodone.mp.	
14. hydromorphone.mp.	
15. levorphanol.mp.	
16. meperidine.mp.	
17. methadone.mp.	
18. morphine.mp.	
19. nalbuphine.mp.	
20. normethadone.mp.	
21. oxycodone.mp.	
22. pentazocine.mp.	
23. pethidine.mp.	
24. propoxyphene.mp.	
25. remifentanil.mp.	
26. sufentanil.mp.	
27. tapentadol.mp.	
28. tramadol.mp.	
29. or/1-28	
30. emerg*.tw,kf.	
31. emergency medical services/ or emergency service, hospital/ or emergency services, psychiatric/	
32. or/30-31	
33. Drug Therapy/	
34. Drug Dosage Calculations/	
35. exp Drug Prescriptions/	
36. Inappropriate Prescribing/	
37. Practice Patterns, Physicians'/	

© 2022 Daoust R et al. *JAMA Network Open*.

38. prescription*.tw,kf.	
39. prescrib*.tw,kf.	
40. ((drug* or medication*) adj4 (discard* or dispos*)).tw,kf.	
41. or/33-40	
42. 29 and 32 and 41	

EMBASE

1. exp narcotic analgesic agent/	
2. opioid*.mp.	
3. opiate*.mp.	
4. opium*.mp.	
5. narcotic*.mp.	
6. alfentanil.mp.	
7. buprenorphine.mp.	
8. butorphanol.mp.	
9. codeine.mp.	
10. dextropropoxyphene.mp.	
11. fentan?l.mp.	
12. hydrocodone.mp.	
13. hydromorphone.mp.	
14. levorphanol.mp.	
15. meperidine.mp.	
16. methadone.mp.	
17. morphine.mp.	
18. nalbuphine.mp.	
19. normethadone.mp.	
20. oxycodone.mp.	
21. pentazocine.mp.	
22. pethidine.mp.	
23. propoxyphene.mp.	
24. remifentanil.mp.	
25. sufentanil.mp.	
26. tapentadol.mp.	
27. tramadol.mp.	
28. or/1-27	
29. emerg*.tw,kw.	
30. emergency health service/	
31. hospital emergency service/	
32. psychiatric emergency service/	
33. or/29-32	
34. dose calculation/	
35. drug misuse/	
36. prescription/	
37. exp inappropriate prescribing/	

© 2022 Daoust R et al. *JAMA Network Open*.

38. clinical practice/	
39. drug utilization/	
40. prescription*.tw,kw.	
41. prescrib*.tw,kw.	
42. ((drug* or medication*) adj4 (discard* or dispos*)).tw,kw.	
43. (therap* or treatment*).ti,kw.	
44. or/34-43	
45. 28 and 33 and 44	

CENTRAL

(ebmz|All EBM Reviews - Cochrane DSR, ACP Journal Club, DARE, CCA, CCTR, CMR, HTA, and NHSEED)

1. exp Narcotics/	
2. exp Opiate Alkaloids/	
3. opioid*.tw.	
4. opiate*.tw.	
5. opium*.tw.	
6. narcotic*.tw.	
7. alfentanil.tw.	
8. buprenorphine.tw.	
9. butorphanol.tw.	
10. codeine.tw.	
11. dextropropoxyphene.tw.	
12. fentan?l.tw.	
13. hydrocodone.tw.	
14. hydromorphone.tw.	
15. levorphanol.tw.	
16. meperidine.tw.	
17. methadone.tw.	
18. morphine.tw.	
19. nalbuphine.tw.	
20. normethadone.tw.	
21. oxycodone.tw.	
22. pentazocine.tw.	
23. pethidine.tw.	
24. propoxyphene.tw.	
25. remifentanil.tw.	
26. sufentanil.tw.	
27. tapentadol.tw.	
28. tramadol.tw.	
29. or/1-28	
30. emerg*.tw,kw.	

© 2022 Daoust R et al. *JAMA Network Open*.

31. Drug Therapy/	
32. Drug Dosage Calculations/	
33. exp Prescription Drug Misuse/	
34. exp Drug Prescriptions/	
35. Inappropriate Prescribing/	
36. Practice Patterns, Physicians'/	
37. prescription*.ti,ab,hw.	
38. prescrib*.ti,ab,hw.	
39. ((drug* or medication*) adj4 (discard* or dispos*)).ti,ab,hw.	
40. or/31-39	
41. 29 and 30 and 40	

Psycinfo

Any Field: analgesic drugs OR
Any Field: narcotic drugs OR
Any Field: opioid* OR
Any Field: narcotic* OR
Any Field: opium* OR
Any Field: opiates OR
Any Field: morphin* OR
Any Field: codeine* OR
Any Field: oxycodo* OR
Any Field: hydroco* OR
Any Field: hydromor* OR
Any Field: fentan* OR
Any Field: meperidin* OR
Any Field: tramadol* OR
Any Field: pethidine* OR
Any Field: methadone* OR
Any Field: alfentanil* OR
Any Field: buprenorphine* OR
Any Field: butorphanol* OR
Any Field: dextropropoxyphene* OR
Any Field: levorphanol* OR
Any Field: nalbuphine* OR
Any Field: normethadone* OR
Any Field: pentazocine* OR
Any Field: propoxyphene* OR
Any Field: remifentanil* OR
Any Field: sufentanil* OR
Any Field: tapentadol*

AND
Any Field: drug therapy OR

© 2022 Daoust R et al. *JAMA Network Open*.

Any Field: drug dosages OR
Any Field: prescription drugs OR
Any Field: prescri*

AND
Any Field: emerg*

Cinahl

(MH “Narcotics+”) OR
(MH “Analgesics, Opioid+”) OR
(MH “Narcotic Antagonists+”) OR
TX opioid* OR
TX opiate* OR
TX opium* OR
TX narcotic* OR
TX alfentanil OR
TX buprenorphine OR
TX butorphanol OR
TX codeine OR
TX dextropropoxyphene OR
TX fentan?l OR
TX hydrocodone OR
TX hydromorphone OR
TX levorphanol OR TX meperidine OR
TX methadone OR
TX morphine OR
TX nalbuphine OR
TX normethadone OR
TX oxycodone OR
TX pentazocine OR
TX pethidine OR
TX propoxyphene OR
TX remifentanil OR
TX sufentanil OR
TX tapentadol OR
TX tramadol
AND

(MH “Drug Therapy”) OR
(MH “Prescriptions, Drug”) OR
(MH “Inappropriate Prescribing”) OR
(MH “Prescribing Patterns”) OR

© 2022 Daoust R et al. *JAMA Network Open*.

TI prescription* OR
AU prescription* OR
TI prescrib* OR
AB prescrib*
AND

TX emerg*

ClinicalTrials.gov

Prescription and Opioid and Emergency

Google Scholar

(opioid OR narcotic) AND (prescribing OR prescription) AND (emergency)

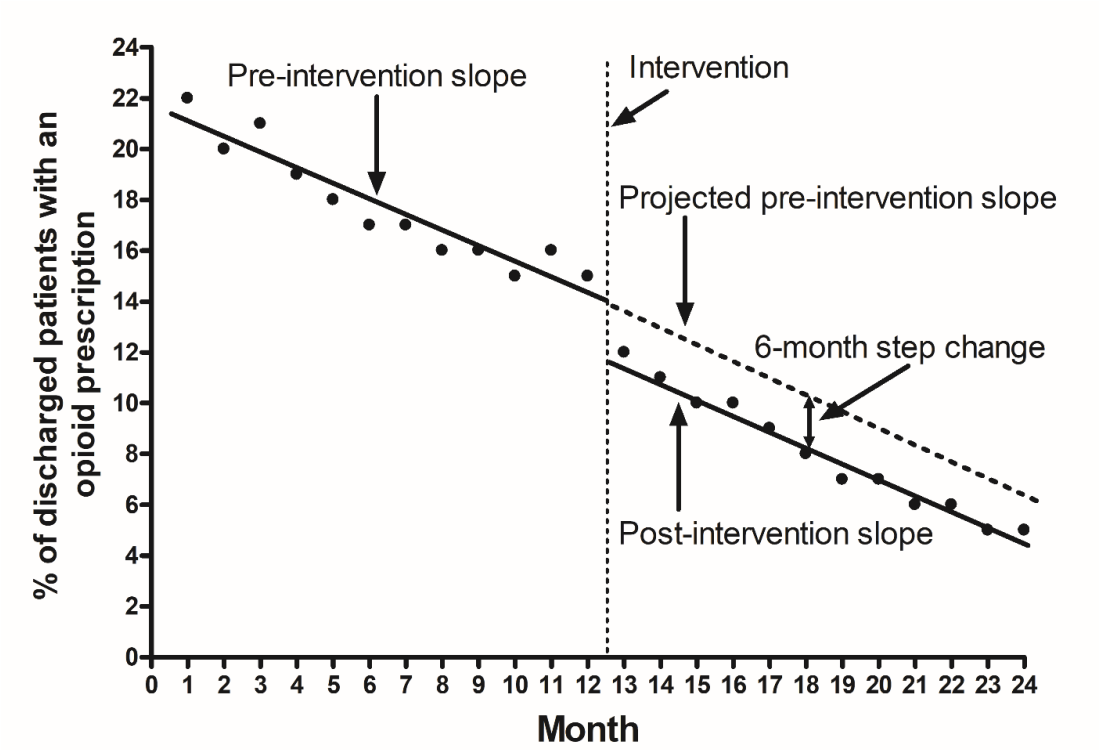
© 2022 Daoust R et al. *JAMA Network Open*.

eMethods 2. Description of the Interventions Design to Reduce Opioid Prescription and How They Were Categorized

Interventions	Detailed description	Category	Rational for categorization
Education	Grand rounds, journal clubs, lectures, conferences, meetings, one-on-one sessions, online modules, courses to opioid prescribers	EPG	Set of information given to change physicians' behavior that are not mandatory
Policy, Guidelines	Documents with guiding decision aims and criteria regarding opioid prescriptions for physicians	EPG	
Clinician Peer Comparison	Comparison/sharing of opioid prescription patterns with other physicians	CPC	Change in physicians' behavior that is influenced by his peers
Prescription Drug Monitoring Program	State prescription drug monitoring program within an electronic database used by physicians to adjust opioid prescribing that keeps track of controlled substance prescriptions	PDMP	Change in physicians' behavior that are mandatory
State Law	State law that specifically modified opioid prescription such as rescheduling of hydrocodone combination products from schedule III to schedule II	PDMP	
Electronic Medical Record Quantity Changes	Modifications of the electronic medical record system by the local healthcare organization to reduce or remove the default quantity of opioids to prescribe	EMR_QC	Change in healthcare organization to modify physicians' opioid prescribing
Physical Therapy	Other approach not directly aimed at physician opioid prescribing	PT	Physicians are not the main target of intervention

© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 1. Example of the 6-mo Step Changes Representation From Interrupted Time Series Data



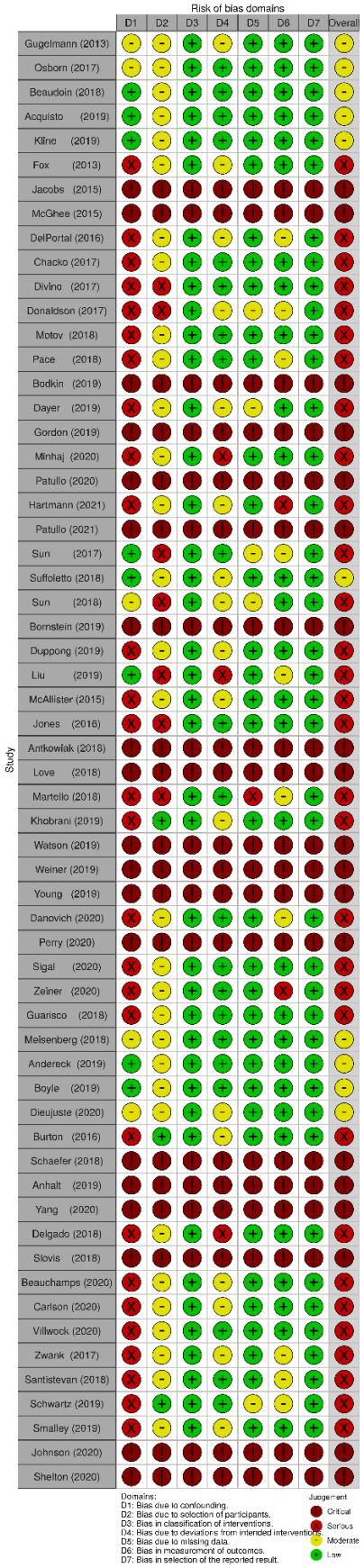
ITS standardizing approach

For time series data that were presented only in graphs, two independent raters (JP and MM) used WebPlotDigitizer (https://automeris.io/WebPlotDigitizer/) to extract data from plots. Intraclass correlation coefficients (ICC) were used to assess interrater reliability. Average data extracted from the graphs were used for ITS analysis when interrater reliability was excellent (ICC>90).[1]

To produce an intervention effect that integrated both the step (immediate) and slope (sustained) changes in ITS studies (if at least three data points before and three data points after the intervention were available), we used segmented time-series regression analyses while taking into account secular time trends and autocorrelation[2] to calculate the 6-month step change.[3] It is calculated as the difference between the level 6-month after intervention on the post-intervention fitted slope and the level predicted by the pre-intervention time trend (eFigure 1 in the supplement). The Durbin-Watson test was used before the segmented time series to assess autocorrelation and, if positive, was added to the model. Since follow-up times varied in ITS studies and the optimal follow-up time is not known, a consensus was obtained across 6 reviewers (RD, JP, MM, JMC, DW, AC) that the 6-month step change would represent a good estimate of the combined effect of immediate and sustained change due to the intervention. For those studies that had fewer than 6 months of post-intervention follow-up, we selected the last available follow-up time point.

Since included ITS studies were not all on the same scale (some used proportion of patients with an opioid prescription and others used crude number of patients), the effect sizes could not be compared directly. Therefore, we calculated a relative 6-month step change by dividing the coefficients and the standard errors by the average of baseline data points and multiplying by 100.[2]

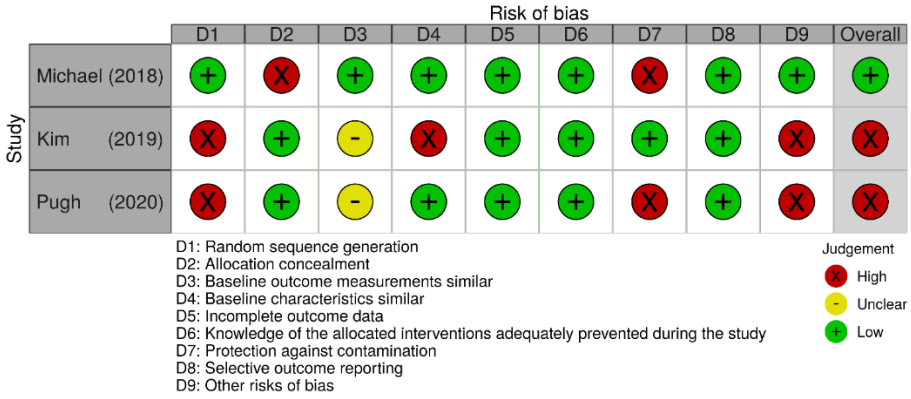
© 2022 Daoust R et al. *JAMA Network Open*.



eFigure 2. Risk of Bias Assessed With the ROBINS-I Tool for ITS and Preintervention-Postintervention Study Designs

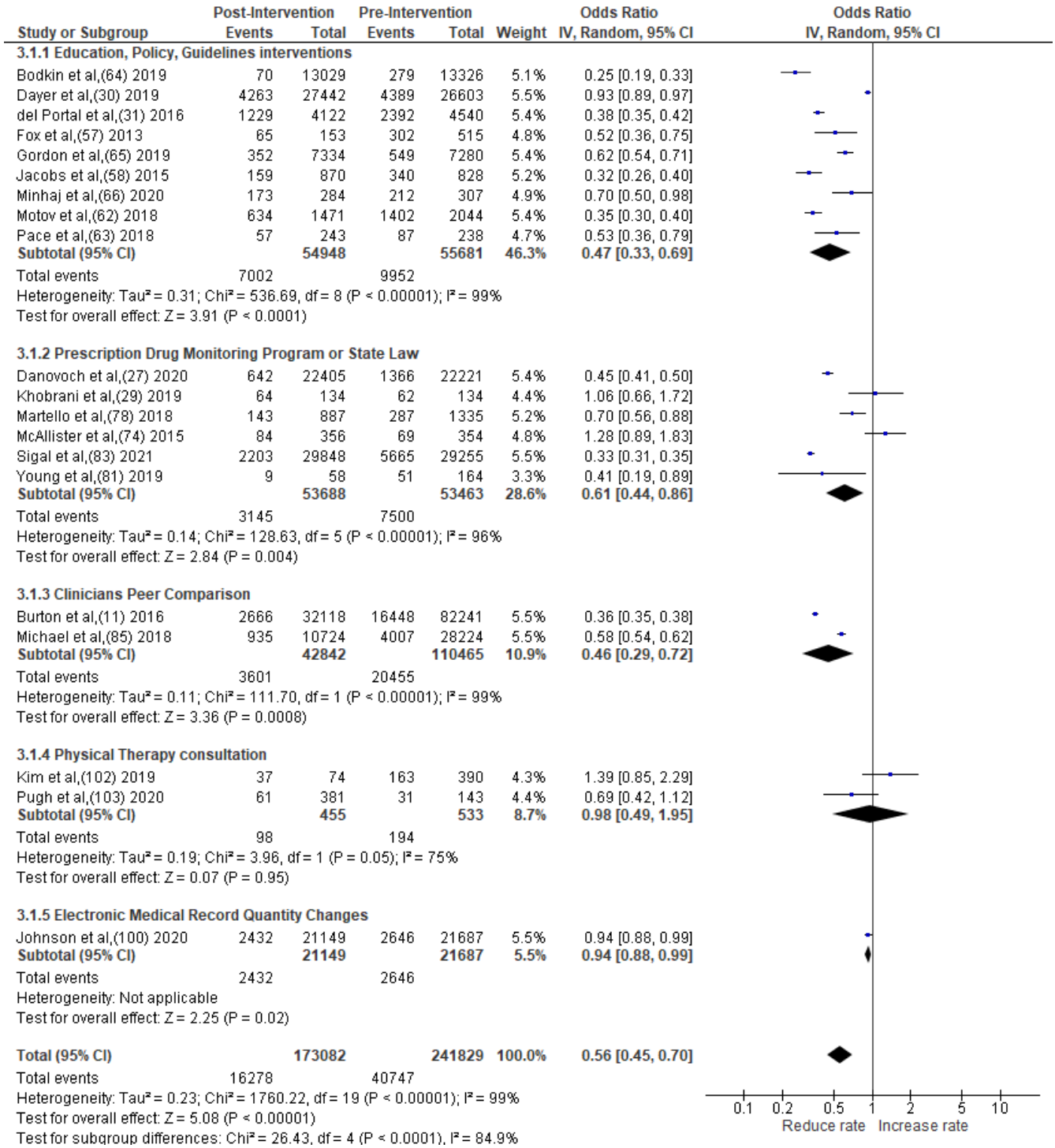
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 3. Risk of Bias Assessed With the EPOC Risk of Bias Tool for the RCT and Cohort Study Designs



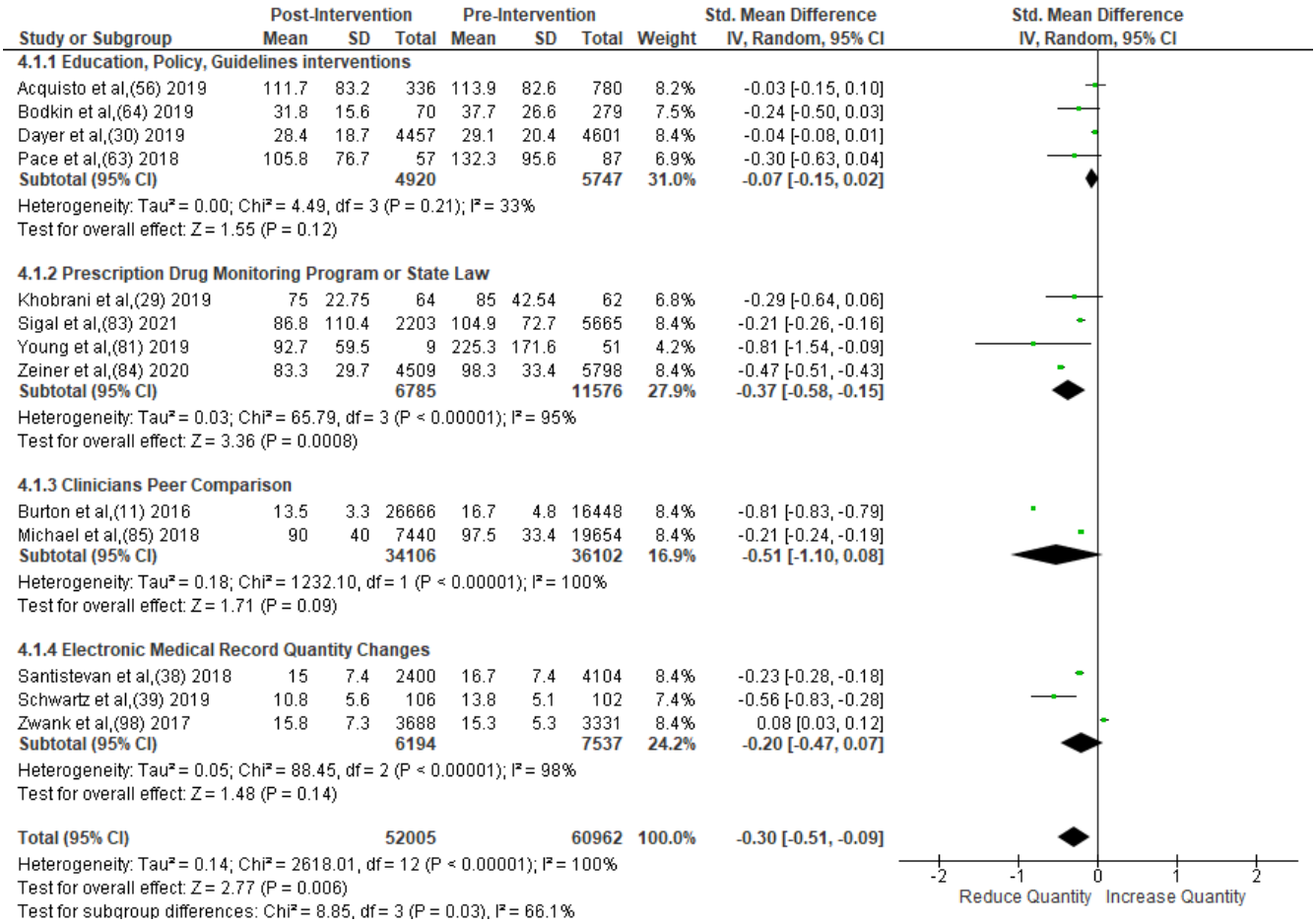
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 4. Forest Plots of Opioid Prescription Rate in RCT, Preintervention-Postintervention, and Cohort Studies



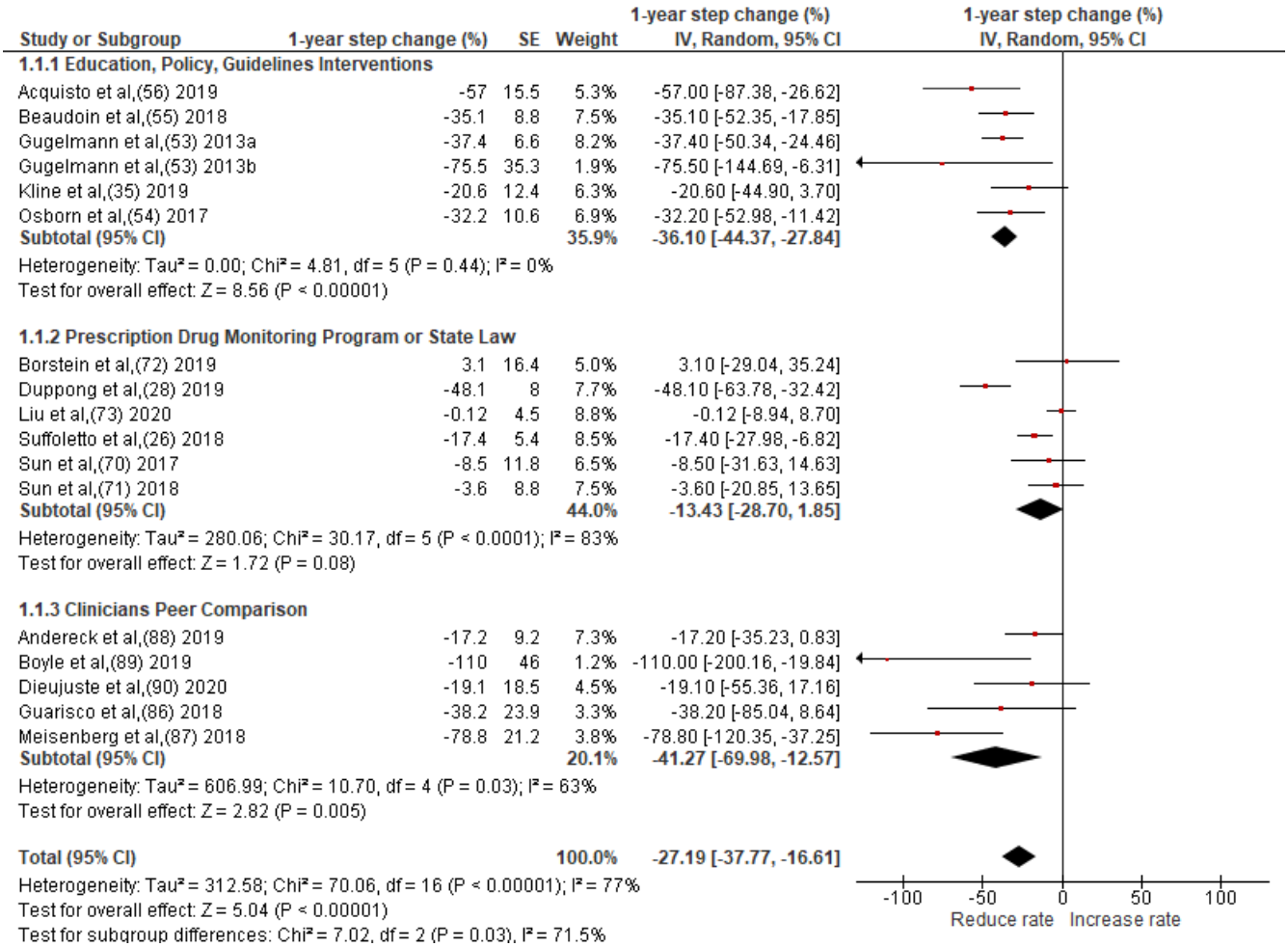
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 5. Forest Plots of Opioid Prescription Quantity in RCT, Preintervention-Postintervention, and Cohort Studies



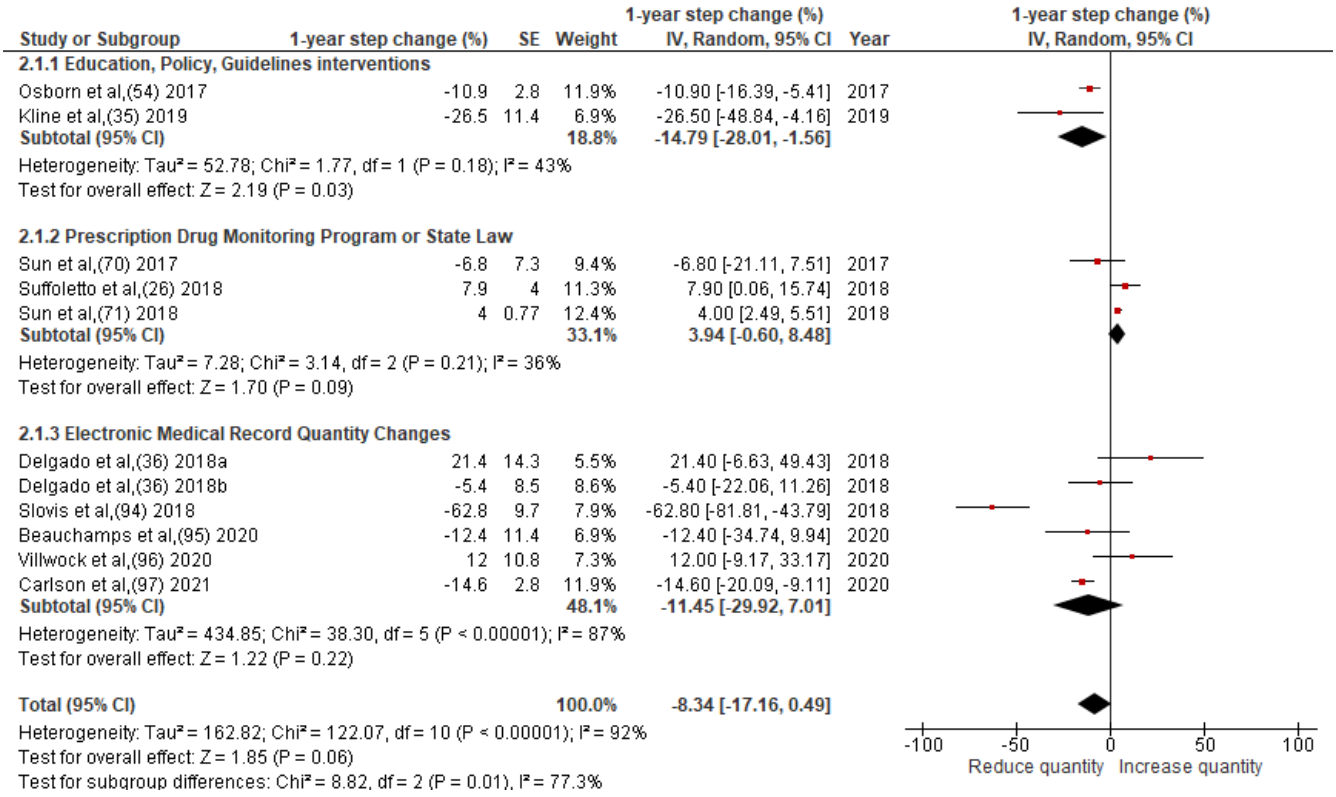
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 6. Forest Plot of the 1-y Step Change in Opioid Prescription Rate for ITS Studies



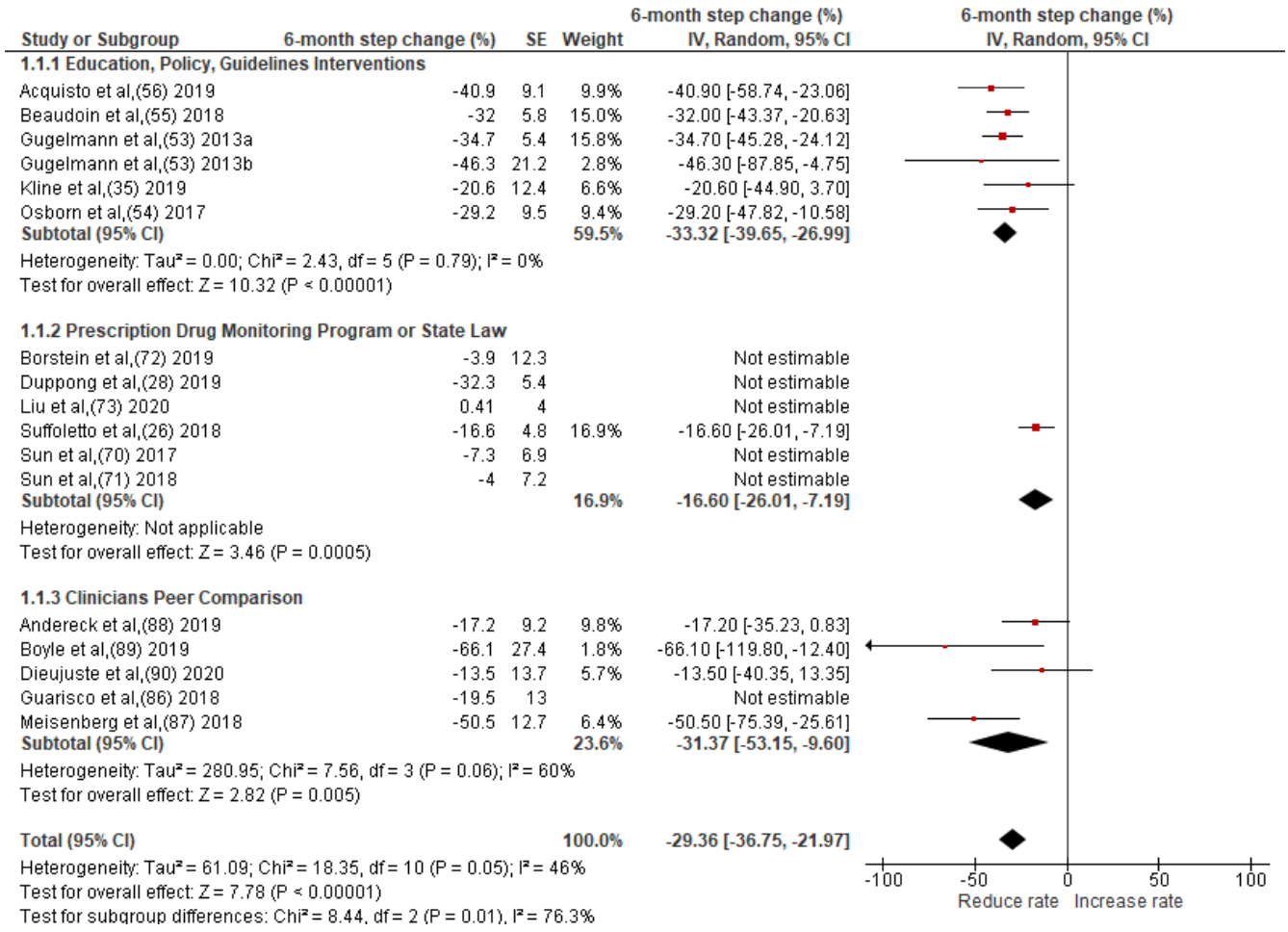
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 7. Forest Plot of the 1-y Step Change in Prescribed Opioid Quantity for ITS Studies



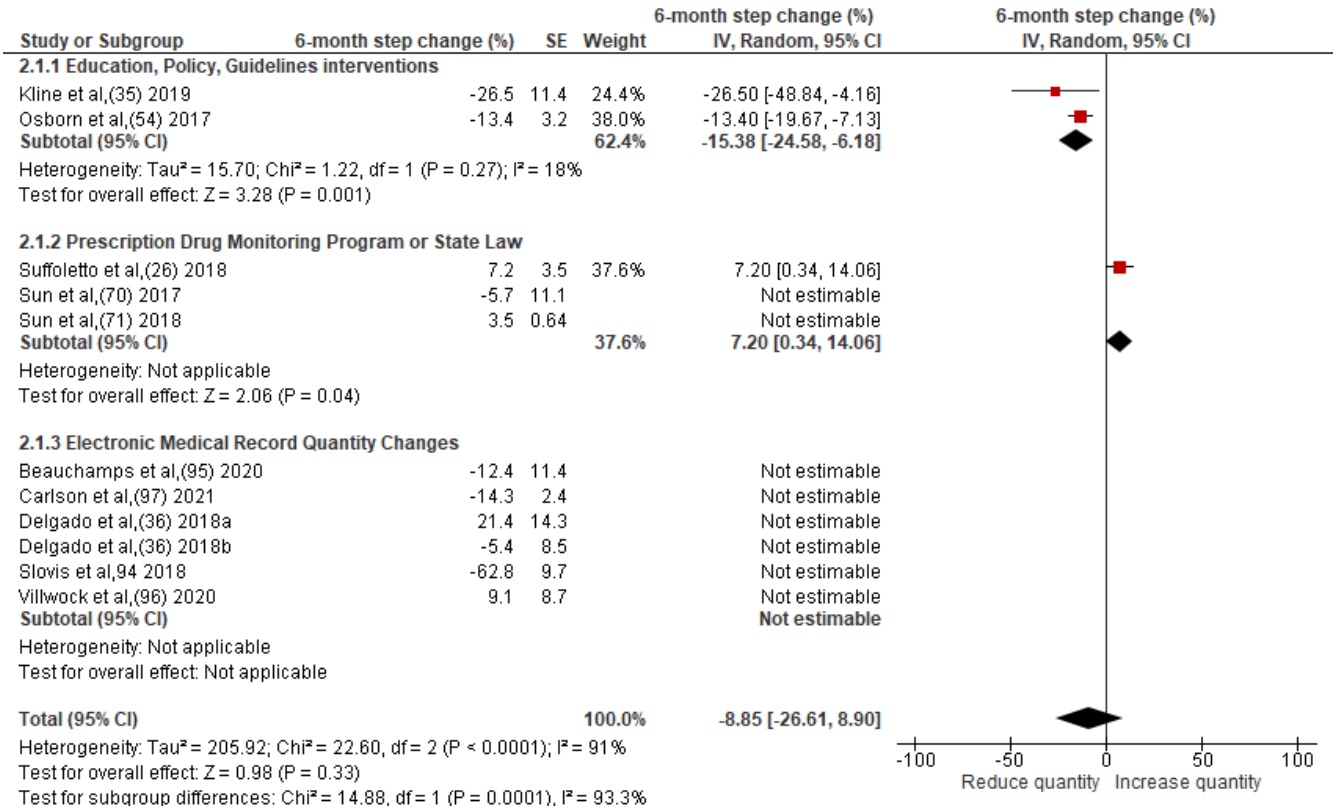
© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 8. Forest Plot of the 6-mo Step Change in Opioid Prescription Rate for ITS Studies Excluding Studies at High Risk of Bias



© 2022 Daoust R et al. *JAMA Network Open*.

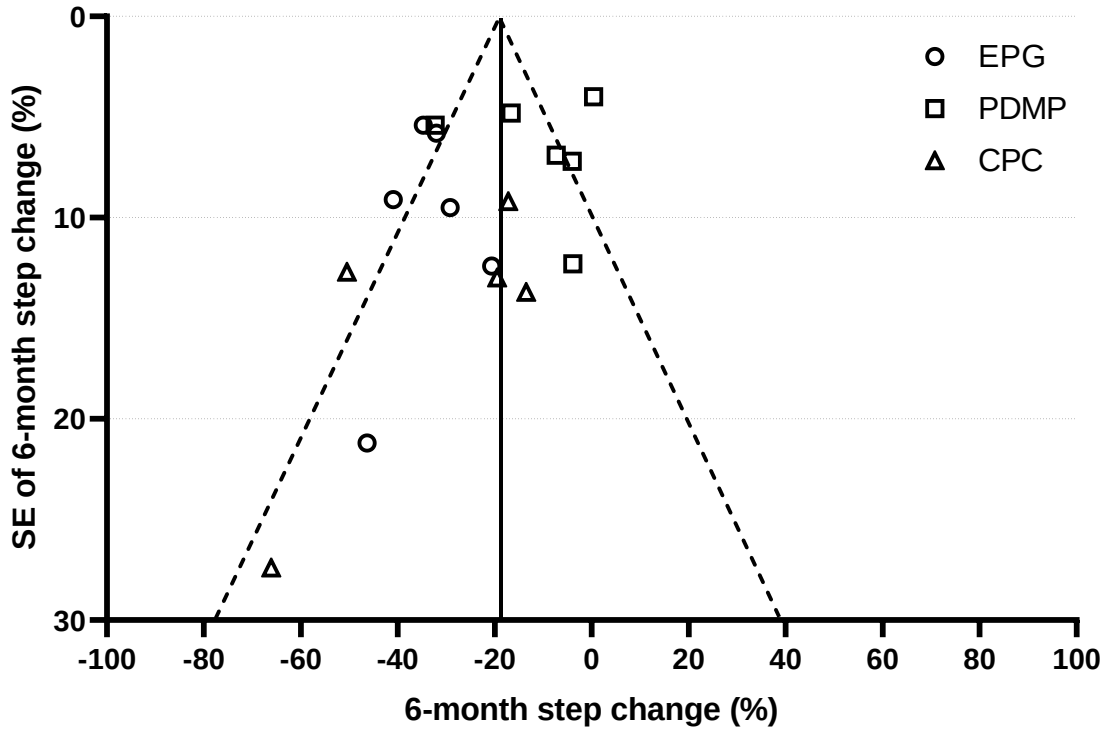
eFigure 9. Forest Plot of the 6-mo Step Change in Prescribed Opioid Quantity for ITS Studies Excluding Studies at High Risk of Bias



© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 10. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Opioid Prescription Rate for ITS Studies

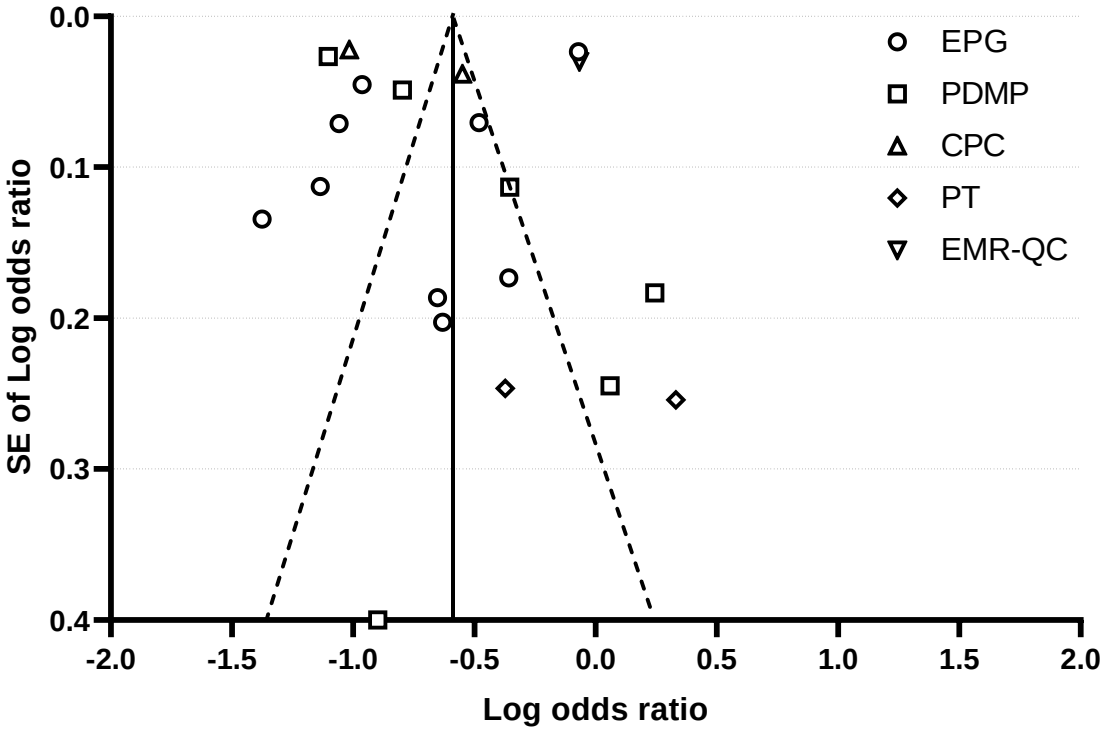
(Egger regression intercept: -1.65; SE: 1.14; p=0.17). EPG: Education, Policy, or Guidelines evaluation; PDMP: Prescription Drug Monitoring Program or State Law; CPC: Clinician Peer Comparison opioid prescribing



© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 11. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Opioid Prescription Rate for RCT, Preintervention-Postintervention, and Cohort Studies

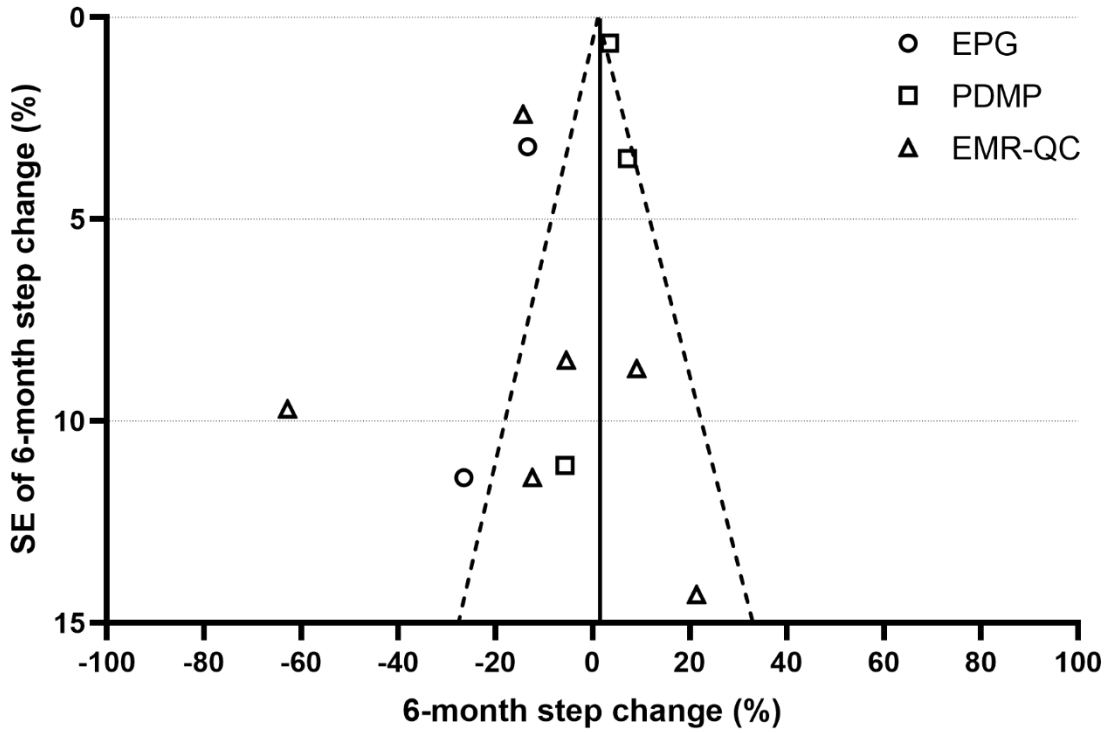
(Egger regression intercept: 0.24; SE: 3.4; p=0.94). EPG: Education, Policy, or Guidelines evaluation; PDMP: Prescription Drug Monitoring Program or State Law; CPC: Clinician Peer Comparison opioid prescribing; PT: Physical Therapy; EMR_QC: Electronic Medical Record Quantity Changes



© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 12. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Opioid Prescription Rate in the Prescribed Opioid Quantity for ITS Studies

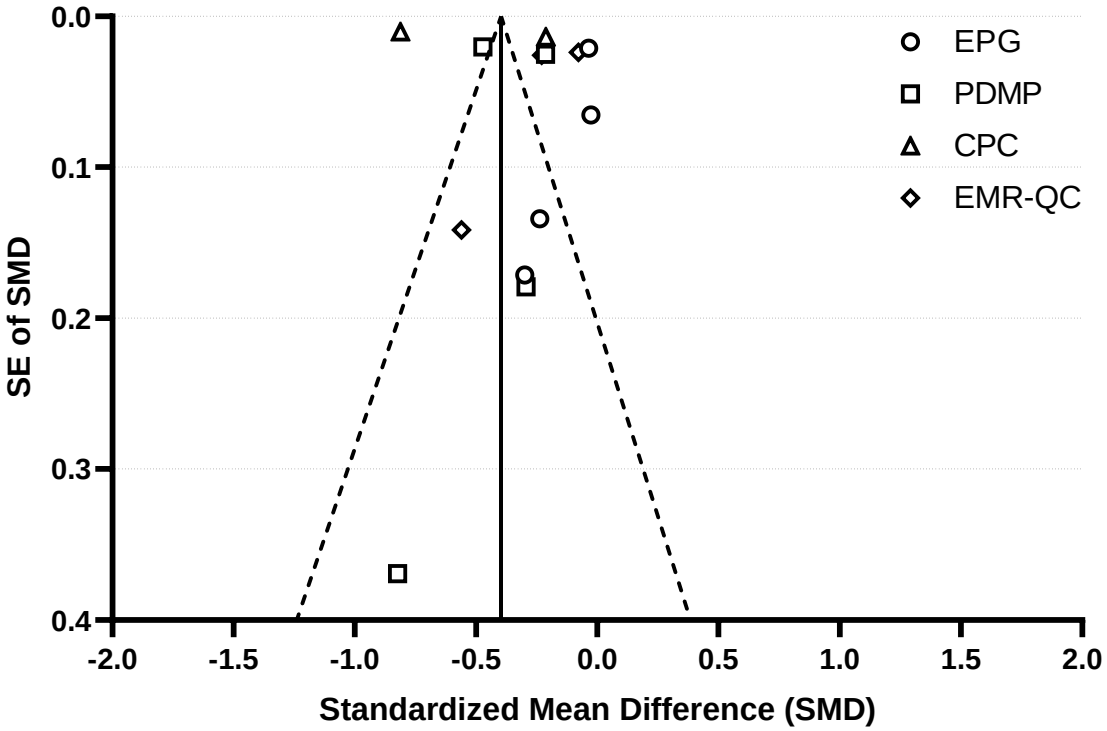
(Egger regression intercept: -2.22; SE: 1.22; p=0.10). EPG: Education, Policy, or Guidelines evaluation; PDMP: Prescription Drug Monitoring Program or State Law; EMR_QC: Electronic Medical Record Quantity Changes



© 2022 Daoust R et al. *JAMA Network Open*.

eFigure 13. Funnel Plot With Pseudo 95% CIs of the 6-mo Step Change in the Prescribed Opioid Quantity for RCT and Preintervention-Postintervention Studies

(Egger regression intercept: 7.08; SE: 5.92; p=0.26). EPG: Education, Policy, or Guidelines evaluation; PDMP: Prescription Drug Monitoring Program or State Law; CPC: Clinician Peer Comparison opioid prescribing; EMR_QC: Electronic Medical Record Quantity Changes



© 2022 Daoust R et al. *JAMA Network Open*.

eReferences

1. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15(2):155-163.
2. Cochrane Effective Practice and Organisation of Care (EPOC). Interrupted time series (ITS) analyses. EPOC resources for review authors, 2017. Accessed December 15, 2020. https://epoc.cochrane.org/resources/epoc-specific-resources-review-authors.
3. Garcia-Elorrio E, Rowe SY, Teijeiro ME, Ciapponi A, Rowe AK. The effectiveness of the quality improvement collaborative strategy in low- and middle-income countries: a systematic review and meta-analysis. *PLoS One*. 2019;14(10):e0221919.

© 2022 Daoust R et al. *JAMA Network Open*.